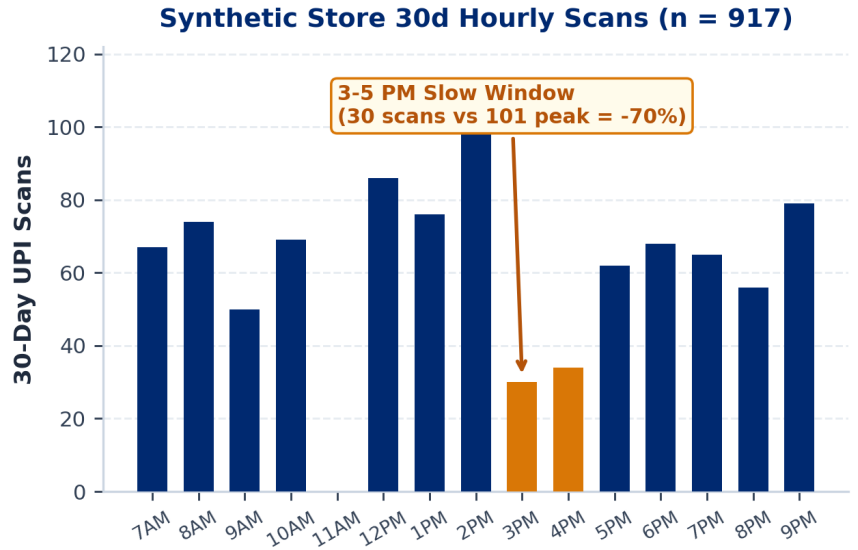
1.64 Cr Merchants Pay for Subscriptions; Retention Stays Manual

- **Reported Paytm Scale:** 1.64 Cr subscription merchants (+27L YoY), Rs 7.1L Cr GMV, Rs 2,648 Cr revenue (Reported: Paytm Q2 FY27 release, 26 Oct 2026).
- **Silent Repeat Churn:** Small shops lose an estimated 25% to 35% of repeat buyers yearly without customer-level alerts (Assumption).
- **Dashboard Friction:** Single-counter operators work 12+ hr shifts; SMB BI dashboards see <8% weekly usage (Assumption).
- **Slow Hours & Rush Prep:** In our synthetic Bengaluru ledger, 3 to 5 PM scans fall 70% below peak while weekend rushes risk stockouts.



PaytmAI VyaparVoice (Track 1)

8:30 PM closing-time voice retention & ops copilot for Soundbox + Paytm for Business.

Team SemiTechnical | Kevin Joseph

One 8:30 PM Voice Action, Governed by Deterministic SQL Rules

- **Plain Voice Brief:** EN author-checked, native review pending; KN/HI draft; zero jargon.
- **Synthetic Store Slice:** Sri Krishna Darshini (240 payers, 1,613 scans); flags 48 regulars silent 16 to 38d.
- **Policy & 80/20 Split:** Filters to 40 DPDP & 14d-cooldown clear buyers: 32 nudged vs 8 silent control.
- **Passes vs Spend Stop:** All 32 nudged get Rs 25 pass (valid while budget lasts); hard stop at Rs 300 (>10% alert).
- **Zero LLM Math:** SQLite computes every rupee and cap; per-visitor cloud sandboxes block paid calls.

TRY LIVE SANDBOX DEMO

https://paytmai-vyaparvoice.onrender.com

Cloud Safeguards Active:
Isolated SQLite per visitor (30m TTL, 64-session LRU, 2MB/32MB disk caps).
Zero paid API calls on public URL.

Deterministic Core & Keyless Caps Built; Rail Hooks Scoped

- **[BUILT] ITT Stats & Atomic Cap Engine:** Fisher exact p, Wilson 95% CIs, pooled/unequal power math, atomic BEGIN IMMEDIATE spend stop, 25-thread concurrency test.
- **[BUILT] Merchant Copy & Cloud Sandboxes:** Merchant vs Technical View, banned-jargon lint, Evidence Grades A/B/C, TTL/LRU/byte-bounded visitor SQLite sessions.
- **[BUILT KEYLESS / PENDING LIVE] Sarvam & Caps:** Keyless Bulbul TTS + Saaras STT adapter, mock-server selftest, SHA-256 audio cache, 15%/15% (30% max) fail-closed credit ledger.
- **[PRODUCTION PLAN] Paytm Ecosystem Hooks:** Paytm QR dynamic coupon settlement, 4G Soundbox push audio, WhatsApp Cloud supplier POs, Postgres/Redis.

Automated Release Guards

Pre-release checks scan for secret keys, banned merchant jargon, and 21 QA assertions.

Atomic Spend-Stop Lock

SQLite write locking stops simultaneous QR scans at the exact merchant rupee cap.

Bounded Visitor Sandboxes

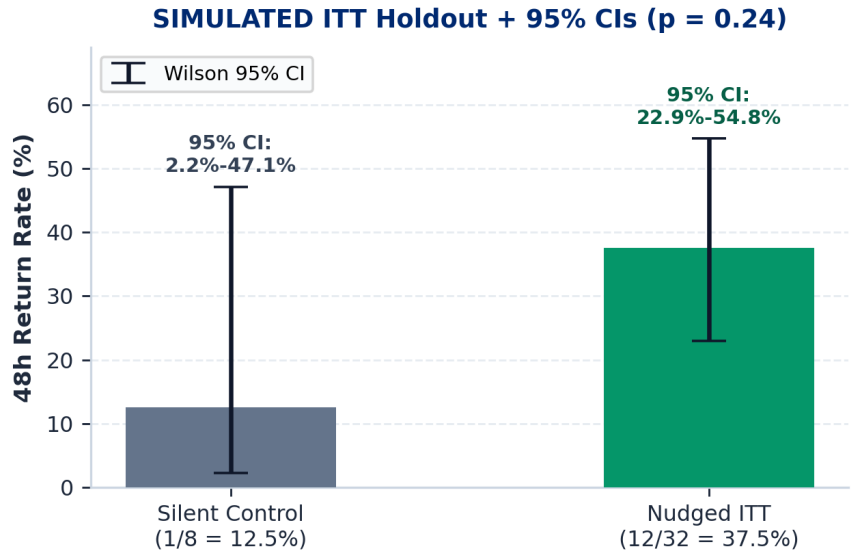
Per-visitor SQLite files enforce 30-min TTL, 64-session LRU eviction, and 32 MB disk cap.

PaytmAI VyaparVoice | Team SemiTechnical (Kevin Joseph)

Intention-to-Treat Holdout, Power & Merchant Net Profit

- **Simulated ITT Return:** 12/32 nudged (37.5%) vs 1/8 control (12.5%); Fisher p = 0.24; simulated 12 equals the Rs 300 cap (0 refused).
- **Sample Requirement:** Single wave (N=40) is directional; 25 pp lift needs 31C + 124T = 155 (~4 store-waves) or 46/arm at 50/50.
- **Sales vs Subsidy:** 8 net visits = Rs 1,740 sales (5.8x sales/promo); 4 organic returners absorb Rs 100 cashback.
- **Merchant Net Profit:** At 30% margin: Rs 522 GP (1.74x); +Rs 222 pre-fee (+74%), +Rs 91.50 after Rs 130.50 fee.
- **5% Fee Base:** Rs 130.50 = 5% of all Rs 2,610 redeemed bills (12 bills, including organic returners).

Intention-to-Treat (ITT) Definition: Compares all 32 assigned nudged buyers vs all 8 assigned silent controls regardless of redemption or cap stops, preventing selection bias and keeping lift invariant if the cap binds.



CAP & BREAK-EVEN DISCLOSURE (SIMULATED)

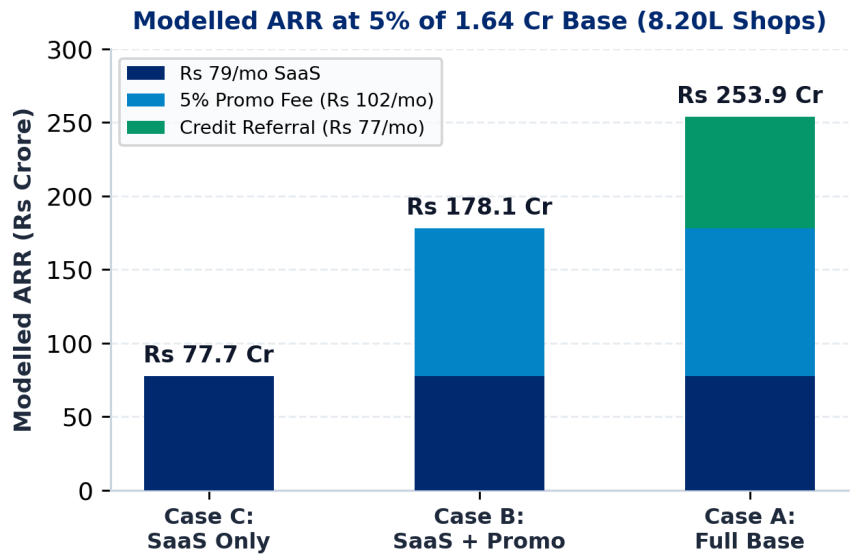
Simulated 12 returners = 12 x Rs 25 = Rs 300 hard spend stop (42.2% overrun risk at p=37.5% triggers >10% alert; suggests Rs 400 cap).

Break-even margin: 17.2% pre-fee | 24.7% post-fee.

Break-even lift (30% margin): +7.8 pp pre | +15.3 pp post.

Three Revenue Cases (8.20L Shops) & 50-Shop Pooled Pilot

- **Case C (SaaS Only):** Rs 79/mo Soundbox add-on x 8.20L shops = Rs 77.7 Cr ARR (zero promo/credit fees).
- **Case B (Core Excl. Credit):** +Rs 102/mo steady promo fee (~0.78 waves/mo, Rs 2,040 redeemed bills) = Rs 178.1 Cr ARR.
- **Case A (Full Base):** +Rs 77/mo credit referral (~7% loan pen. Assumption) = Rs 253.9 Cr ARR.
- **Unit COGS & Margin:** Rs 48/mo cached (Rs 5 AI + Rs 43 msg; 81.4% margin) vs Rs 71.80 uncached TTS (72.2%).
- **Pooled 50-Shop Pilot:** 50 shops x 1 wave (~400C + ~1,600T, stratified by shop): 83.3% power at +6 pp, 97.2% at +8 pp.



Why Pool 50 Shops (Not 4 Waves/Shop): 14d cooldown & ~0.78 waves/mo replenishment preclude 4 weekly waves in 1 shop; pooling 50x1 gives 50.3%/68.8%/83.3%/92.5%/97.2%/99.8% power at +4/+5/+6/+7/+8/+10 pp.

Q2 FY27 BASE & FEE RECONCILIATION

5% of 1.64 Cr subscription merchants (Reported: Paytm Q2 FY27, 26 Oct 2026) = 8.20L shops.

1 backlog wave = Rs 130.50 fee; steady state (~0.78 waves/mo) = Rs 102/mo.

APPENDIX A1 | TRACK 1: MERCHANT GROWTH AI (UPDATED TO PAYTM Q2 FY27)

PaytmAI VyaparVoice

Closing-Time Voice Growth, Operations & Retention Copilot

Built for Paytm's 1.64 Cr subscription-paying merchant deployments (Reported: Paytm Q2 FY27 release, 26 Oct 2026). Combines deterministic SQL segmentation, ITT spend-cap accounting, and plain-language voice briefings.

01 : Plain Voice + Button

EN author-checked, native review pending; KN/HI draft

02 : ITT Cap-vs-Holdout

All 32 nudged get passes; Rs 300 hard spend stop & >10% risk alert

03 : 50-Shop Pooled Pilot

+Rs 91.50 post-fee; 400C+1,600T gives 83.3% power at +6 pp lift

TEAM: SemiTechnical | Member: Kevin Joseph | Live: paytmai-vyaparvoice.onrender.com

Deck v2.1 Status: Stage 1 verified (stage1-ok); pooled 50-shop power, ITT definition, and copy disclosures applied.

Problem Details, Primary Paytm Q2 FY27 Sources & Synthetic Persona

1. Silent Repeat Churn (25% to 35% Yearly - ASSUMPTION)

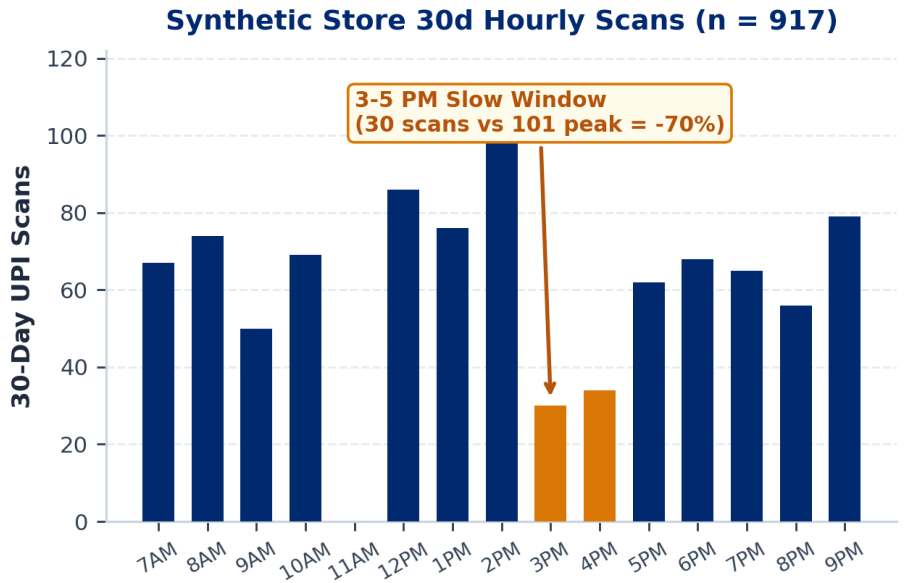
Merchants hear isolated payment alerts without customer-level retention tracking, losing repeat buyers before cashflow drops are noticed.

2. Dashboard Friction (<8% WAU - ASSUMPTION)

Counter-operated kirana, darshini, and pharmacy shops lack time to write prompts, export CSVs, or configure CRM rules.

3. Slow Hours & Rush Prep (SYNTHETIC STORE)

In our 60-day synthetic Indiranagar ledger, 3 to 5 PM scans fall 70% below peak while weekend morning rushes risk staple stockouts.



- **Reported (Paytm Q2 FY27, 26 Oct 2026):** 1.64 Cr subscription-paying merchants, Rs 7.1L Cr merchant GMV, Rs 2,648 Cr operating revenue.
- **Assumptions vs Synthetic:** ~7% merchant loan pen. & Rs 90-100/mo Soundbox fee (Assumptions); Ramesh Rao store is synthetic (917 scans/30d).

Track 1 Coverage: Growth, Operations, and Customer Recovery

PILLAR 1 : GROW BUSINESS

Retention & Slow-Hour Copilot

- **Churn Detection (Built):** SQL flags regulars silent 16+ days (16-38d in synthetic DB).
- **Slow-Hour Group (Built):** Identifies 3-5 PM snack buyers (55 detected, 34 eligible).
- **ITT Spend Stop (Built):** Issues passes to all nudged buyers; atomic cap stops spend.
- **QR Coupon Hook (Planned):** Production hook into Paytm QR redemption rail.

PILLAR 2 : MANAGE OPERATIONS

Rush Prep & Credit Signals

- **Rush Alert (Simulated):** Flags weekend morning demand spikes from scan history.
- **Supplier PO (Simulated):** Dispatches pre-filled WhatsApp restock (+18L milk, +12kg batter).
- **Dual Ledgers (Built):** Separates merchant cashback stop from 30% Sarvam credit cap.
- **Credit Signal (Planned):** Feeds verified cashflow stability to partner NBFC funnel.

PILLAR 3 : SERVE CUSTOMERS

Privacy-Safe Service Recovery

- **Delay Tagging (Simulated):** Flags peak-hour queue delays and 3-star feedback tags.
- **Goodwill Pass (Simulated):** 1-tap bilingual apology + Rs 30 next-scan coffee voucher.
- **Zero PII Leakage (Built):** No raw phone numbers or payer IDs ever leave SQLite.
- **DPDP & Cooldown (Built):** Hard opt-in filter and 14-day per-payer anti-spam cooldown.

Intention-to-Treat Cap-vs-Holdout & Late-Arrival Handling

STEP 1 : PASS ISSUANCE

Filter 48 -> 40 eligible (80/20: 32 nudged, 8 control). Issue Rs 25 passes ('valid while shop offer budget lasts') to ALL 32 nudged buyers (Rs 800 face).

STEP 2 : >10% & 80% ALERTS

At approval, P(Binomial(32, 0.375) > 12) = 42.2% (>10% alert -> suggests Rs 400 cap). Live at 80% cap (10/12 paid), prompt merchant 1-tap top-up.

STEP 3 : INDEPENDENT DRAW

48h simulator draws customer return visits independently of max_budget first, then applies atomic BEGIN IMMEDIATE stop at Rs 300 (12 paid).

STEP 4 : ITT & DECLINE LOG

After cap, normal UPI clears, cashback withholds, refused_redemptions + audit_log increment (brand risk tracked). ITT (32 vs 8) stays unchanged.

DISCLOSED BRAND RISK & VERIFIED INVARIANCE TESTS

- **Post-Cap Customer Experience & Brand Risk:** Pass terms state 'valid while shop offer budget lasts'; declining a 13th regular still risks counter disappointment, so overrun alerts, 80%-cap top-up prompts, and logged declines are mandatory.
- **Automated Concurrency & ITT Tests:** 25-thread simultaneous scan test verifies budget_spent == Rs 300 (12 paid, 13 refused); multi-seed test proves ITT holdout comparison is identical whether cap is Rs 800 or Rs 100.

Component Status: Built vs Pending vs Production Plan

TOUCHPOINTS	[BUILT] Merchant & Technical View UI	[BUILT] 1-Tap & Voice Chip Fallback	[PLANNED] Soundbox 4G Push & WhatsApp Cloud
VOICE & LLM	[BUILT] Browser Speech + SHA-256 Cache	[PENDING LIVE] Keyless Sarvam (30% Cap)	[STAGE 2] Keyless Groq Grounded English Q&A
CORE ENGINE	[BUILT] FastAPI + TTL/LRU/Byte Sessions	[BUILT] 5-Rule Guardrails + Jargon Lint	[BUILT] ITT Fisher/Wilson/Power + Profit ROI
DATA & OPS	[BUILT] Atomic SQLite Spend-Stop Ledger	[BUILT] Automated Preflight + Stress Suite	[PLANNED] PostgreSQL & Redis Cluster

Differentiation vs Manual Khata Apps and Generic Chatbots

Dimension	Manual BI / Khata Apps	Generic LLM Chatbots	PaytmAI VyaparVoice
Merchant UX	Charts & manual ledger entry	Requires typing business prompts	**8:30 PM plain voice brief + 1-tap button**
Customer Graph	Incomplete manual phonebooks	No link to payment history	**Paytm QR repeat-payer IDs (synthetic in MVP)**
Financial Safety	Manual coupon rules	LLM can invent unsafe offers	**Deterministic floor, atomic spend stop & >10% alert**
Privacy (DPDP)	Raw phone lists exposed	Prompt data leakage risk	**SQL opt-in filter; zero PII sent to voice/LLM**
Causal Proof	Vanity message counts	No comparison group	**80/20 ITT holdout + 50-shop stratified pilot**

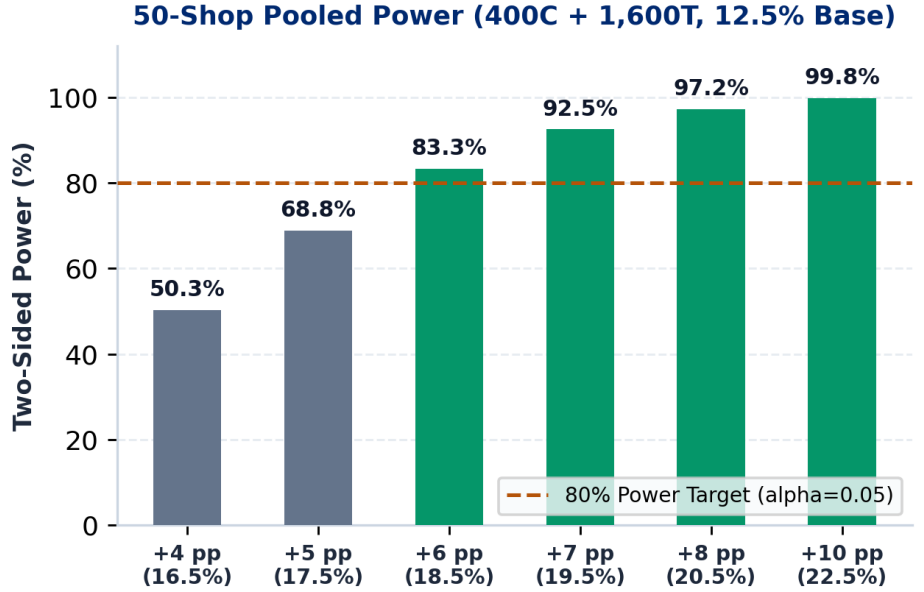
50-Shop Stratified Power (400C + 1,600T) & Pilot Targets

POOLED 50-SHOP X 1-WAVE POWER (12.5% BASELINE, ALPHA=0.05)

- **Why Pool Stores:** 50 shops x 1 wave (~400C + ~1,600T, stratified by shop); avoids claiming 1 shop runs 4 waves in 4w under a 14d cooldown (~0.78 waves/mo).
- **Exact Power Curve:** +4 pp: 50.3% | +5 pp: 68.8% | +6 pp: 83.3% | +7 pp: 92.5% | +8 pp: 97.2% | +10 pp: 99.8%.
- **Break-Even Lift:** +7.8 pp pre-fee (20.3%) and +15.3 pp post-5%-fee (27.8% at 30% shop margin; >99.9% power).

ECOSYSTEM HYPOTHESES (UNVALIDATED PILOT TARGETS)

- **Repeat & Waste Targets:** +14% to +22% repeat visits; -18% afternoon perishable loss (PILOT TARGETS).
- **Device & QR Share:** -35% Soundbox churn; +12% to +15% store Paytm QR GMV lift (PILOT TARGETS).
- **Credit Funnel:** Expand ~7% merchant loan pen. (ASSUMPTION) toward 15% (LONG-TERM TARGET).



83.3%

Power at +6 pp Lift (400C + 1,600T)

+15.3 pp

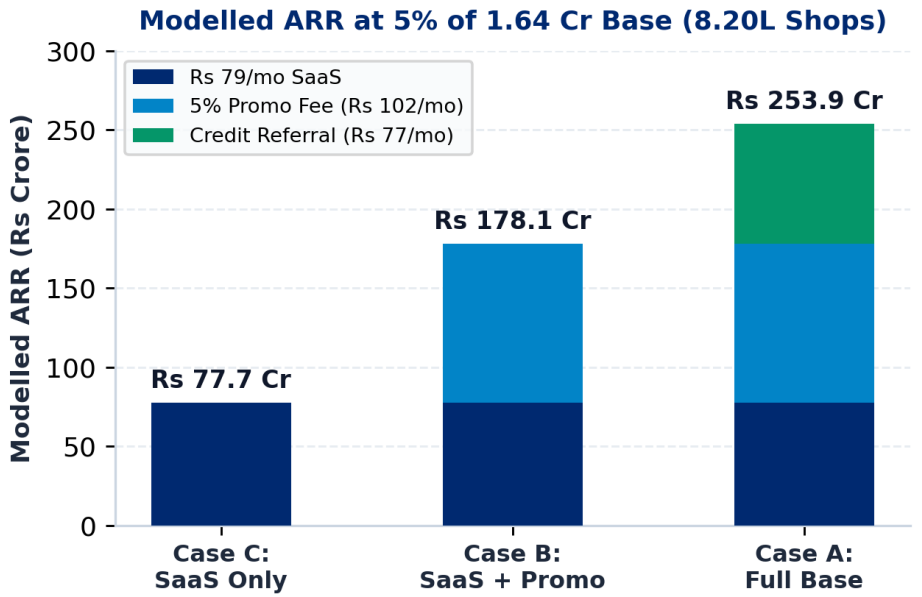
Post-Fee Break-Even (+7.8 pp Pre-Fee)

Rs 5.00

Cached AI Cost/Mo (Rs 28.80 Uncached)

Q2 FY27 ARR Cases, Fee Reconciliation & TTS Cache Sensitivity

- **Stream 1 (SaaS - Rs 79/mo):** Add-on to Soundbox (~Rs 90-100/mo rental ASSUMPTION) x 8.20L shops (5% of 1.64 Cr Q2 FY27 base) = Rs 77.7 Cr ARR.
- **Stream 2 (Promo Fee - Rs 102/mo):** 5% fee on ALL redeemed bills incl. organic returners: 1 wave = Rs 130.50 (Rs 2,610 bills); steady state assumes ~0.78 waves/mo (Rs 2,040/mo bills) = Rs 102/mo (+Rs 100.4 Cr -> Rs 178.1 Cr ARR).
- **Stream 3 (Credit - Rs 77/mo):** Amortized 1.5%-2.5% partner origination uplift (~7% loan pen. ASSUMPTION) -> +Rs 75.8 Cr (Rs 253.9 Cr ARR).
- **Cached COGS (Rs 48.00/mo):** Rs 1.80 SLM + Rs 3.20 TTS (~88% phrase cache hit on Bulbul v3) + Rs 43 msg = 81.4% gross margin.
- **Uncached COGS (Rs 71.80/mo):** Uncached Bulbul v3 (9k chars at Rs 30/10k = Rs 27 TTS) = 72.2% margin (39.2% cached margin on SaaS-only).



FEE & BASE RECONCILIATION

1. Fee Base: 5% applies to gross QR-redeemed promo bills including organic returners.

2. Cadence: Initial backlog wave yields Rs 130.50; steady replenishment (~0.78 waves/mo) yields Rs 102/mo.

3. Q2 FY27 Base: 5% x 1.64 Cr = 8.20L shops.

APPENDIX A9 : VALIDATION & GO / NO-GO DECISION RULE

From Verified Synthetic MVP to Pooled 50-Shop Stratified Pilot

STAGE A : BUILT & VERIFIED

Deterministic Core & Safeguards

- FastAPI + TTL/LRU/byte-bounded SQLite sessions (204 KB/visitor)
- ITT cap-vs-holdout, >10% alert, 25-thread concurrency & preflight
- Keyless Sarvam mock selftest (30% cap) + banned-jargon lint

STAGE B : POOLED 50-SHOP PILOT

Stratified Single-Wave Validation

- 3-merchant counter teach-back across EN (author-checked), KN & HI (draft)
- 50 shops x 1 wave (~400C + ~1,600T, stratified by shop; 83.3% power at +6 pp)
- Advance only if lower 95% bound of post-5%-fee net profit/nudged > 0 AND lift > break-even (+15.3 pp)

STAGE C : PRODUCTION SCALE

Paytm Ecosystem Integration

- Bind Paytm QR dynamic coupon settlement & 4G Soundbox push
- Enforce live 80%-cap merchant top-up prompt to protect counter goodwill
- Scale across 8.20L target stores (Rs 77.7 Cr to Rs 253.9 Cr ARR)

Merchant Plain-Language View vs Technical Audit View

MERCHANT VIEW (ZERO JARGON - LINT VERIFIED)

- **Plain Outcome Copy:** Early check on one store wave (pool multiple shops to be sure): 12 of 32 with offer came back vs 1 of 8 without.
- **Clear Spend Stop:** Your 300 rupee limit was reached after 12 paid redemptions (0 extra declined in base run).
- **Estimated Shop Profit:** At 30% shop margin, estimated profit is Rs 222 before Paytm fee (Rs 91.50 after 5% fee).
- **Honest Copy Status:** EN author-checked, native review pending; KN & HI carry visible [DRAFT - UNREVIEWED] badges.

TECHNICAL & AUDIT VIEW (ANALYST / JUDGE)

- **ITT & Wilson 95% CIs:** 37.5% [22.9%, 54.8%] vs 12.5% [2.2%, 47.1%]; two-sided Fisher exact p = 0.24.
- **Pooled Pilot Power:** 50 shops x 1 wave (400C + 1,600T): 50.3%, 68.8%, 83.3%, 92.5%, 97.2%, 99.8% at +4..+10 pp.
- **Cap Risk & Refusals:** Face exposure Rs 800; P(X > 12 | p=0.375) = 42.2% (>10% alert -> suggests Rs 400 cap).
- **Banned Merchant Terms:** Automated regex blocks cohort, holdout, p-value, Fisher, Wilson, ROI, DPDP, RFM, ITT.

GRADE A : OBSERVED LEDGER

SQLite store counts (240 payers, 1,613 scans, 48 silent -> 40 eligible; synthetic seed 20261025).

GRADE B : SIMULATED ITT

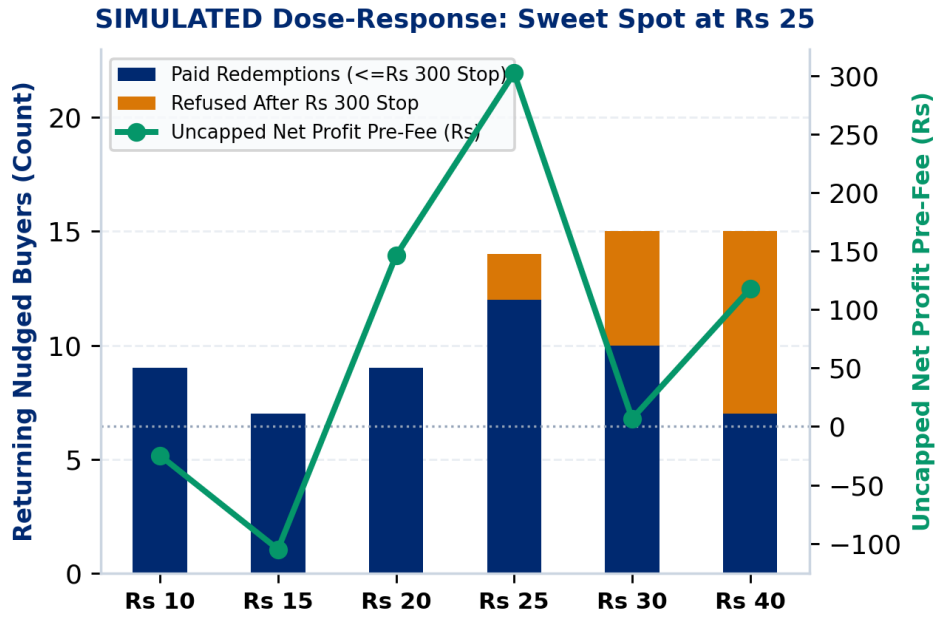
Randomized 32 vs 8 split, independent outcome draw, Fisher p & Wilson CIs (simulated).

GRADE C : ASSUMPTIONS

30% shop gross margin, 5% fee on redeemed bills, ~0.78 waves/mo steady cadence.

Merchant Spend Stop, Sarvam Credit Cap & Dose-Response Curve

- **Merchant Cashback Stop:** Issues passes ('valid while budget lasts') to all 32 nudged buyers; SQLite lock stops spend at Rs 300 (12 paid) and logs declines.
- **Dose-Response Sweet Spot:** In simulation, Rs 25 peaks uncapped profit: offers of Rs 30 to 40 erode uncapped net profit via organic subsidy AND trigger 5 to 8 cap refusals under a Rs 300 stop.
- **Margin & Lift Break-Even:** At +25 pp simulated lift, shop breaks even at 17.2% margin pre-fee (24.7% post-fee); at 30% margin, break-even lift is +7.8 pp pre-fee (+15.3 pp post-fee).
- **Sarvam 30% Credit Ledger:** Separate JSON ledger caps TTS <=15% and STT <=15% (30% total of 500 sponsor credits = 150 max) and halts fail-closed on HTTP 402/429/5xx.
- **Public Cloud Lock:** PUBLIC_DEPLOY=true (auto-enabled on Render) hard-blocks live paid calls while SHA-256 audio cache replays at 0 credits.



DUAL CAP SUMMARY (SIMULATED & BUILT)

1. Merchant Cap: Protects shopkeeper pocket (Rs 300 stop; >10% overrun warning & 80% top-up prompt).

2. Sponsor API Cap: Protects hackathon credits (<=15% TTS, <=15% STT, fail-closed).